Appending CSV Data Sources in Power BI and Using Folder Connector

1. Introduction

In real-world business scenarios, it is very common to receive data in multiple files that have the same structure but represent different periods, regions, or categories. These files often need to be combined so that the analysis can be conducted on a unified dataset. Power BI provides a set of powerful features in Power Query to enable such data combination and transformation tasks.

It extends that understanding by demonstrating how to append *multiple CSV files*, manage mismatched column headers, and finally how to combine many files automatically using the **Folder Connector**.

These techniques are extremely useful because organizations frequently store data in separate CSV files for various years, product categories, or geographies, and analysts need to bring them together into a single dataset for reporting.

YATICORP
EMPOWERING EDUCATION

---

2. Starting the Appending Operation Using CSV Files

2.1 Creating a New Blank Report

The first step is to open a fresh Power BI Desktop file. Clicking **Blank Report** initializes a clean Power BI workspace. Power BI loads its interface and prepares the canvas for new data connections. This setup phase may take a few seconds depending on system performance.

Once the blank PBIX file is ready, the next step is to connect to the data sources we want to append.

---

3. Connecting to CSV Data Sources

3.1 Using the Text/CSV Connector

Power BI provides a dedicated connector for CSV data.
To use it:

1. Click **Get Data**.
2. Choose **Text/CSV** from the list of common data sources.
3. Browse and select the first CSV file — here, **TourismData_Asia.csv**.

Power BI reads the file and presents a **preview**. In many CSV files, Power BI may not correctly detect column headers. This is common when the CSV file does not explicitly specify the header row.

In the preview of the Asia dataset, we may observe:

- Column headers appear as **Column1, Column2, Column3**, etc.
- Actual headers are present in the **first row** of data.

YATICORP
EMPOWERING EDUCATION

Thus, for proper data modeling and transformation, the first important step is to **promote the first row to headers**.

---

4. Transforming Data in Power Query

After clicking **Transform Data**, the Power Query Editor opens.

4.1 Using "Use First Row As Headers"

To clean the structure:

- Click **Home → Use First Row as Headers**.

This action replaces generic column names with the actual column headers.

Next, verify the **data types** applied to each column.
Power BI attempts automatic data type detection, but it may need manual correction in some cases. Data types are essential to ensure accurate calculations, aggregations, and visualizations.

---

5. Importing the Second CSV File

Once the first dataset is ready, we import the second CSV file:

1. In Power Query, click **New Source**.
2. Choose **Text/CSV**.
3. Select **TourismData_NonAsia.csv**.

As before, promote the first row to headers and check the data types.

While comparing both tables, observe the critical issue:

- In the Asia dataset, the final column header is **Previous Year**.
- In the Non-Asia dataset, the final column header is **Last Year**.

YATICORP
EMPOWERING EDUCATION

This mismatch will cause problems during appending.

---

6. Understanding Column Mismatch Problems During Appending

6.1 How Power BI Appends Tables

Appending is a **vertical combination** of tables — the rows of table B are placed below the rows of table A.

For appending to work correctly:
✔ **Column headers must match exactly** (case-sensitive, spelling-sensitive).
✔ **The number of columns need NOT be the same**, but mismatched columns lead to null values.

6.2 What Happens When Headers Mismatch?

If we append without fixing mismatched headers:

- Power BI creates **two separate columns**:
    - One named *Previous Year*
    - One named *Last Year*
- The rows coming from the Asia file show values only under *Previous Year* column.
- The rows from the Non-Asia file show values only under *Last Year* column.
- The other column will show **nulls**.

This leads to undesirable data duplication and confusion.

YATICORP
EMPOWERING EDUCATION

7. Demonstrating Appending With Mismatched Headers

To illustrate the issue:

1. Click **Home → Append Queries → Append Queries as New**.
2. Select the two tables:
    - TourismData_Asia
    - TourismData_NonAsia
3. Click **OK**.

Power BI generates a new appended table.

Upon examining the last columns:

- There are two columns representing the same information.
- Values appear only in one column per dataset.

- This produces inconsistent and unusable structure.

Having understood the problem, we now explore the solutions.

---

8. Solution 1 — Fix Mismatched Columns After Appending

One way to resolve the mismatch is to **merge the two columns** after appending:

Steps:

1. Select the two mismatched columns:
    - Previous Year
    - Last Year
2. Click **Transform → Merge Columns**.
3. Choose **Separator: None**.
4. Click **OK**.

YATICORP
EMPOWERING EDUCATION

Power BI merges both values into one combined column.

Rename this merged column as **Previous Year**.

Advantages:

- Quick fix when there are only a few mismatched columns.
- Useful when appending has already been done.

Limitations:

- Not efficient for large datasets.
- Not advisable when there are multiple mismatches.

9. Solution 2 — Fix the Column Names Before Appending (Recommended)

A better method is to **correct the column header mismatch before appending**.

Steps in Non-Asia Table:

1. Double-click the header **Last Year**.
2. Rename it as **Previous Year**.
3. Press Enter.

Now both CSV tables share identical column structures:

- Region
- 2013
- 2014
- 2015
- 2016
- 2017
- Previous Year

Now the append operation will work cleanly.

YATICORP
EMPOWERING EDUCATION

10. Performing the Correct Append Operation

With consistent column names:

1. Click **Home → Append Queries → Append Queries as New**.
2. Select:
    - First Table: TourismData_Asia
    - Second Table: TourismData_NonAsia
3. Click **OK**.

Power BI now produces a perfectly appended table with:

- No null-filled duplicate columns.
- Harmonized structure.

YATICORP
EMPOWERING EDUCATION

Rename this new table as **Appended Table** or **Tourism Combined**.

11. Applying the Queries and Loading the Data

Click **Close & Apply**.
Power BI processes:

- All transformation steps
- All rename operations
- The append operation
- Data load
- Relationship validation

Depending on hardware and background tasks, query execution time may vary.

Once this is complete, both:

- The original tables, and
- The newly appended table
  appear in the **Data pane**.

The file can be saved as **Tourism Appended.pbix**

12. Appending Multiple Files Using Folder Connector

12.1 Real-World Need

Organizations often store monthly, quarterly, or yearly data in dozens—or even hundreds—of separate files. Manually connecting each file individually and appending them is inefficient.

Power BI solves this by enabling users to:
✔ Import **all files in a folder**
✔ Automatically combine them
✔ Apply transformations that replicate across all files
✔ Create a scalable and dynamic data ingestion pipeline

This is one of the most powerful features of Power BI.

YATICORP
EMPOWERING EDUCATION

13. Preparing Files for Folder Import

Suppose we have three files:

- Sales_2022.csv
- Sales_2023.csv
- Sales_2024.csv

To use the folder connector:

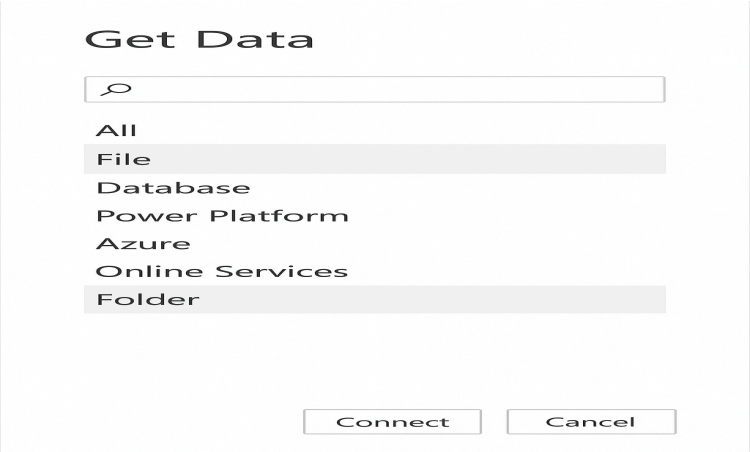
1. Create a folder named **All Sales Data**.
2. Copy all these CSV files into that folder.

14. Connecting Power BI to a Folder

Unlike CSV or Excel connectors, “Folder” is not always available in the quick menu.

Steps:

1. Open a new Blank PBIX file.
2. Click **Get Data → More**.
3. Search or select **Folder**.
4. Click **Connect**.



A dialog box opens asking for the **folder path**.

You may:

- Browse the folder, or
- Copy the folder path from File Explorer and paste it.

YATICORP
EMPOWERING EDUCATION

Click **OK**.

15. Viewing Folder Contents

Power BI now displays:

- File name
- Extension
- Date accessed
- Date created

- Folder path
- Attributes
- Content binary

You now have three options:

1. **Combine & Load**
2. **Combine & Transform (recommended)**
3. **Transform Data**

To apply transformations before loading the data, select:

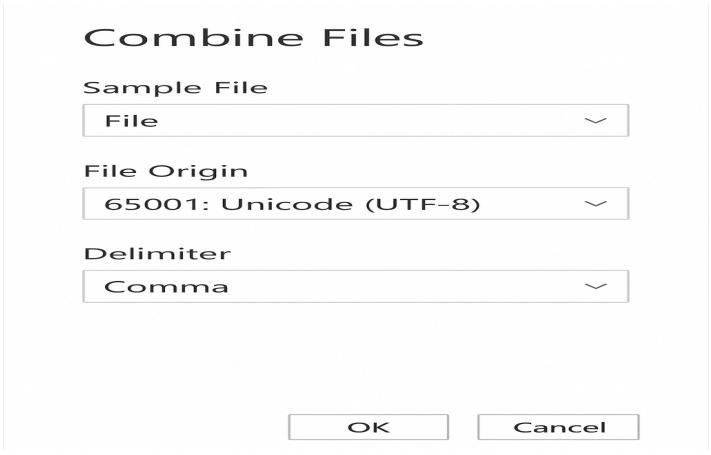
→ **Combine & Transform Data**

YATICORP
EMPOWERING EDUCATION

16. The Combine Files Dialog Box

Power BI opens a configuration window showing:

- Sample file
- Column delimiter
- Encoding
- Metadata extraction options



In most cases, the default settings are sufficient.

Click **OK**.

Power BI performs:

- Automatic sample file analysis

- Query generation
- File iteration through each CSV
- File combination
- Data model construction

Then Power Query Editor opens.

17. Power Query Steps Generated Automatically

When combining files from a folder, Power BI automatically creates several queries:

Helper Queries

- **Transform Sample File**
- **Sample File**
- **Parameter for sample file**
- **Automatically generated helper functions**

YATICORP
EMPOWERING EDUCATION

These are used internally by Power BI but generally do not require editing.

Main Query

A final query (often named **Transform File**) contains the combined data from all CSV files.

---

18. Understanding the Query Logic

Power BI performs the following behind the scenes:

1. Read the folder.
2. Identifies all supported files (CSV in this case).

3. Runs the **Transform File** function on each file.
4. Combines the transformed data.
5. Outputs a unified dataset.

This is extremely efficient because:

- Any newly added files placed in the folder will be included automatically on refresh.
- File-level transformations become reusable templates.

---

19. Applying and Loading Folder-Based Append Queries

Once the combined dataset is ready:

- Click **Close & Apply**.

YATICORP
EMPOWERING EDUCATION

Power BI processes thousands of rows (e.g., 2000+ per file). Depending on the number of files and data volume, it may take a few seconds.

After completion:

- A single combined dataset for **all three years** is visible.
- Helper queries remain hidden unless expanded.

You now have a professionally combined dataset ready for visualization.

---

20. Key Learnings

1. Appending CSV Files Individually

- Use **Text/CSV** connector.
- Promote the first row as headers.

- Fix data types.
- Address header mismatches before appending.

2. Handling Header Mismatches

- Option A: Fix headers before append (recommended).
- Option B: Append first and then merge mismatched columns.

3. Using Append Queries

Two modes:

- **Append Queries** (modifies existing table)
- **Append Queries as New** (creates a new table)

Always use "As New" for preserving raw data.

4. Folder Connector for Many Files

- Best method for appending 10, 20, 100, or more files.
- Dynamically refreshes with new files added to the folder.
- Automates repetitive transformations.

YATICORP
EMPOWERING EDUCATION

21. Conclusion

Appending data is one of the most essential ETL operations performed inside Power BI. Whether you are consolidating department-wise files, yearly transactions, regional performance sheets, or any repeated structure files, the append feature enables seamless merging.

This session covered:

- How to append two CSV files.
- How to handle mismatching headers.
- How to merge columns when mismatches occur.
- How to rename columns before appending.
- When to use Append Queries and Append Queries as New.
- How to combine **multiple files automatically** using the Folder connector.

YATICORP
EMPOWERING EDUCATION

These skills not only help in managing structured datasets but also allow analysts to build scalable and automated reporting systems. With proper use of the Folder Connector, Power BI becomes a powerful tool for recurring data ingestion and transformation.